Rainfall Variability in India
A Statistical Analysis

Deep diving in understanding Indian rainfall in statistical aspects.

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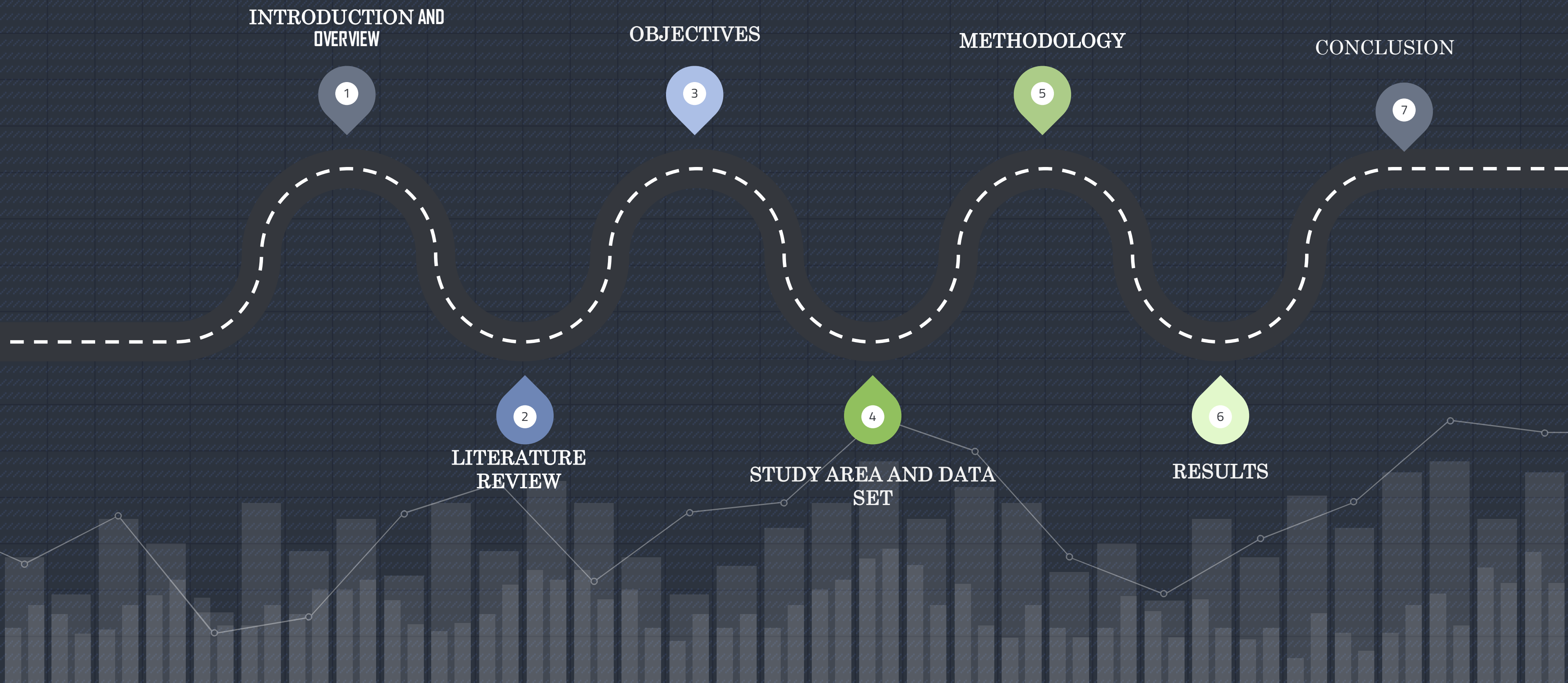
ROADMAP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1) INTRODUCTION AND OVERVIEW

Rainfall Analysis in India :

- Rainfall is a critical component of the Earth's climate system, essential for replenishing water sources, supporting agriculture, and maintaining ecosystems. It plays a key role in regulating temperature and humidity and is vital for hydropower generation. However, excessive rainfall can cause floods, soil erosion, and infrastructure damage, while insufficient rainfall can lead to crop failures and water shortages. Studying rainfall patterns and identifying suitable probability distributions for rainfall data is crucial for predicting extreme weather events, managing water resources, and developing effective mitigation strategies. This presentation explores the importance of rainfall analysis in India, highlighting its benefits and challenges.

- To deepen our comprehension, we explore previous investigations into rainfall analysis carried out in various parts of the world and India.

2) **Literature Review: Prior Studies**

- Northern Pakistan: Amin et al. (2016) found the normal distribution best fit rainfall data at the Marden station.
- River Tiber, Rome: Calenda et al. (2009) proposed an index for optimal distribution considering accuracy and uncertainty.
- United States: Arora et al. (1989) compared methods for log Pearson type 3 distribution, noting performance differences.
- Malaysia: Zalina et al. (2002) identified the generalized extreme value distribution as the most suitable for maximum rainfall estimates.
- Bangladesh: Ghosh et al. (2016) concluded the generalized extreme value distribution best fit most stations.
- Brazil: Ozonur et al. (2021) found the three-parameter lognormal distribution generally provided the best fit.

Rainfall Analysis in India: Recent Studies

- Maharashtra: Gandhre et al. (2024) identified suitable extreme value distributions using the Anderson–Darling test.
- Junagadh Region (Gujarat): Gundalia et al. (2023) found the Gumbel distribution best predicted rainfall maxima.
- India (1871-2005): Kumar et al. (2010) observed significant rainfall increases in Coastal Karnataka, Punjab, and Haryana.
- India (1901-2015): Praveen et al. (2020) analyzed long-term trends, noting a declining trend post-1951.
- Roorkee: Kaur et al. (2021) predicted annual maximum rainfall using various distributions, with the Gumbel distribution proving most effective.
- Uttarakhand: Kumar et al. (2017) found the Weibull distribution best suited for annual rainfall data in 46% of districts.

Rainfall Analysis in India: Recent Studies

- Chhattisgarh: Mandal et al. (2015) provided insights into managing climate variability impacts through data downscaling.
- Sagar Island: Mandal et al. (2015) analyzed severe rainfall-induced climate risks, identifying distributions suited for different seasons.
- Kelantan River Basin: Ng et al. (2021) fitted probability distributions to data for flood mitigation efforts.

3) OBJECTIVES

- We will be using the data collected from annual maximum rainfall data for 36 meteorological regions.

- From the literature review, we will be identifying the best fit probability distributions.

- The Maximum Likelihood Estimator will be used to identify the parameters of all the probability distributions selected.

- We will be using four Goodness-of-fit tests i.e. K-S, A-D, CVM and BIC to identify the best-fit probability distributions for each region.

- We will be using the Ranking method to identify the best fit probability distribution.

- We will be analyzing the best-fit probability distributions identified in the literature review.
- On the basis of best fit probability distribution, we will then determine the return periods for 5, 10, 25, 50, 100 and 200 years across all the regions.

- In our project, we are conducting a detailed examination of India's rainfall patterns utilizing extensive data provided by the Indian Meteorological Department. Our study involves segmenting this data into 36 distinct regions based on geographical criteria. We then employ various probability distribution models, including both conventional ones like Normal (N2) and Log-Normal (LN2), as well as specialized models such as Pearson Type III (PIII3), Gumbel (GUM2), Generalized Extreme Value (GEV3), Weibull (W2), Generalized Pareto (GP3), and Pearson Type V (PV3) distributions.
- Through rigorous testing methodologies such as Kolmogorov-Smirnov (KS), Anderson-Darling (AD), Cramer-von Mises (CVM), and Bayesian Information Criterion (BIC) tests, we aim to assess the accuracy and reliability of these models. Additionally, we utilize visual tools like Quantile-Quantile (Q-Q) plots to further validate our findings.
- By analysing the data over different time intervals, ranging from 5 to 200 years, we seek to uncover insights into extreme rainfall events and long-term trends. Ultimately, our goal is to provide valuable insights to policymakers, researchers, and stakeholders to support informed decision-making in the management of India's water resources.

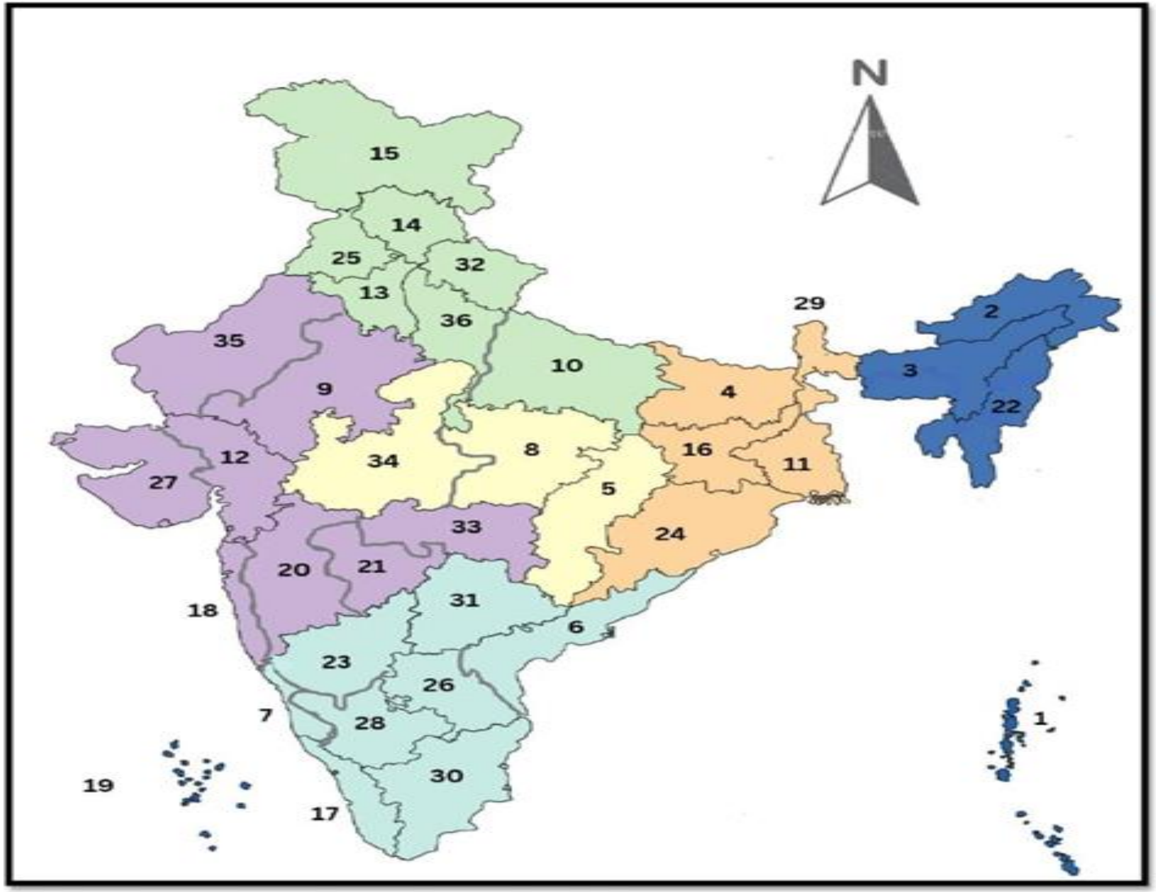
4) Study Area and Data Set

India's Diverse Climate:

- Four distinct seasons: cold weather, hot weather, advancing monsoon, and retreating monsoon. Varied topography influences regional climatic patterns.
- Cold Weather Season (Winter):
  November to February.
  Low temperatures, clear skies, occasional frost, and snowfall in Himalayan regions.
- Hot Weather Season (Summer):
  March to May.
  Rising temperatures, 'loo' winds in northern India, and pre-monsoon mango showers.
- Advancing Monsoon (Rainy Season):
  June to September.
  Arrival of Southwest monsoon winds, widespread rainfall vital for agriculture and water resources.
- Retreating Monsoon (Transition Season):
  October to November
  Gradual withdrawal of monsoon, transitioning to cooler and drier conditions.

Data Collection:

- 121 years of data (1901-2022).
- Annual maximum rainfall recorded across 36 meteorological regions.
- Data sources: data.gov.in (1901-2015) and hydro.imd.gov.in (2016-2022).



1. Andaman and Nicobar Island
2. Arunachal Pradesh
3. Assam and Meghalaya
4. Bihar
5. Chattisgarh
6. Coastal Andhra Pradesh
7. Coastal Karnataka
8. East Madhya Pradesh
9. East Rajasthan
10. East Uttar Pradesh
11. Gangetic West Bengal
12. Gujarat
13. Haryana , Chandigarh & Delhi
14. Himachal Pradesh
15. Jammu and Kashmir and Ladakh
16. Jharkhand
17. Kerala
18. Konkan and Goa
19. Lakshadweep
20. Madhya Maharashtra
21. Marathwada
22. Nagaland, Manipur, Mizoram and Tripura
23. North Interior Karnataka
24. Orissa
25. Punjab
26. Rayalaseema
27. Saurashtra ,Kutch and Diu
28. South Interior Karnataka
29. Sub-Himalayan West Bengal and Sikkim
30. Tamil Nadu and Pondicherry
31. Telangana
32. Uttaranchal
33. Vidarbha
34. West Madhya Pradesh
35. West Rajasthan
36. West Uttar Pradesh

5) METHODOLOGY

- In this study, we have used 9 best-fit probability distribution based on Goodness-of-fit tests i.e. K-S, AD, CVM and BIC on 36 regions for extreme rainfall analysis and have been presented below along with their probability density functions, cumulative distribution functions and their parameters: -

Table1. Probability Distributions

Distribution	Probability Density Function (pdf)	Cumulative Distribution Function (cdf)	Domain	Parameter
N2	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\left[-\frac{1}{2\sigma^2}(x-\mu)^2\right]}$	$F(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\exp\left[-\frac{1}{2\sigma^2}(x-\mu)^2\right]\right) dx$	$-\infty < x < \infty$	$-\infty < \mu < \infty$, $0 < \sigma < \infty$ where $\mu = mean$, $\sigma = sd$
LN2	$f(x) = \frac{1}{x\sigma_Y\sqrt{2\pi}} \exp\left[-\frac{1}{2\sigma_Y^2}(ln(x)-\mu_Y)^2\right]$	$F(x) = \frac{1}{\sigma_Y\sqrt{2\pi}} \int_0^x \left(\frac{1}{x}\exp\left[-\frac{1}{2\sigma_Y^2}(ln(x)-\mu_Y)^2\right]\right) dx$	$x \in (0,\infty)$	$\mu \in R$, $\sigma \in (0,\infty)$ μ =shape parameter σ =scale parameter
PIII3	$f(x) = \frac{1}{\|\alpha\|\Gamma\beta}\left[\left(\frac{x-\xi}{\alpha}\right)^{\beta-1}\right]\exp\left[-\left(\frac{x-\xi}{\alpha}\right)\right]$	$F(x) = \frac{1}{\|\alpha\|\Gamma\beta}\int_{\xi}^{x}\left[\left(\frac{x-\xi}{\alpha}\right)^{\beta-1}\right] * \exp\left[-\left(\frac{x-\xi}{\alpha}\right)\right] dx$	$0 \le x \le \infty$	$\alpha > 0$, $\beta > 0$ where α = scale parameter β = shape parameter ξ =location parameter
LPIII3	$f(x) = \frac{1}{\|\alpha\|x\Gamma\beta}\left[\left(\frac{\ln x-\xi}{\alpha}\right)^{\beta-1}\right]\exp\left[-\left(\frac{(ln(x)-\xi)}{\alpha}\right)\right]$	$F(x) = \frac{1}{\|\alpha\|\Gamma\beta}\int_0^x \frac{1}{x}\left(\frac{(\ln(x)-\xi)}{\alpha}\right)^{\beta-1}\exp\left[-\left(\frac{(ln(x)-\xi)}{\alpha}\right)\right] dx$	$0 < x \le e^{\xi}$, $\alpha < 0$ $e^{\xi} \le x < \infty$, $\alpha > 0$	$\alpha > 0$, $\beta \neq 0$ where α = scale parameter β = shape parameter ξ =location parameter
PV3	$f(x) = \frac{e^{-\frac{\beta}{x-\gamma}}}{\beta\Gamma\alpha\left(\frac{x-\gamma}{\beta}\right)^{\alpha+1}}$	$F(x) = \int_{\gamma}^{\infty}\left(\frac{e^{-\frac{\beta}{x-\gamma}}}{\beta\Gamma\alpha\left(\frac{x-\gamma}{\beta}\right)^{\alpha+1}}\right) dx$	$\gamma < x < \infty$	$\alpha > 0$, $\beta > 0$, $\gamma \neq 0$ where α = shape parameter β = scale parameter γ =location parameter
GUM2	$f(x) = \left(\frac{1}{\alpha}\right)\exp\left[-\left(\frac{x-\beta}{\alpha}\right) - exp\left(-\left(\frac{x-\beta}{\alpha}\right)\right)\right]$	$F(x) = \exp\left[-exp\left(-\frac{x-\beta}{\alpha}\right)\right]$	$-\infty \le x \le \infty$	$\beta \in R$, $\alpha > 0$ where α = scale parameter

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				β = location parameter
GEV3	$f(x)=\frac{1}{\beta}\left(1+\gamma\left(\frac{x-\alpha}{\beta}\right)\right)^{-\left(1+\frac{1}{\gamma}\right)}$ $*\exp\left(-\left(1+\gamma\left(\frac{x-\alpha}{\beta}\right)\right)^{-\frac{1}{\gamma}}\right)$	$F(x)=\exp\left\{-\left[1+\gamma\left(\frac{x-\alpha}{\beta}\right)\right]^{-\frac{1}{\gamma}}\right\}, if\ \gamma\neq 0$ $=\exp\left\{-exp\left[-\left(\frac{x-\alpha}{\beta}\right)\right]\right\},\ \gamma=0$	$x>\alpha-\frac{\beta}{\gamma},\gamma>0$ $x<\alpha-\frac{\beta}{\gamma},\gamma<0$	$-\infty<\alpha<\infty,\quad \beta>0$ $-\infty<\gamma<\infty$ where α= location parameter $\beta = scale\ parameter$ γ= shape parameter
W2	$f(x)=\left(\frac{k}{\alpha}\right)\left(\frac{x}{\alpha}\right)^{k-1}\exp\left\{exp\left(-\left(\frac{x}{\alpha}\right)^{k}\right)\right\}$	$F(x)=1-\exp\left\{exp\left(-\left(\frac{x}{\alpha}\right)^{k}\right)\right\}$	$x>0$	$\alpha>0,\quad k>0$ α = shape parameter k = scale parameter
GP3	$f(x)=\frac{1}{\sigma}\left(1-k\left(\frac{x-\xi}{\sigma}\right)\right)^{\frac{1}{k}-1},\quad k\neq 0$ $=\frac{1}{\sigma}\exp\left(-\frac{x-\xi}{\sigma}\right),\quad k=0$	$F(x)=1-\left(1-k\left(\frac{x-\xi}{\sigma}\right)^{\frac{1}{k}}\right), k\neq 0$ $=1-\exp\left(-\frac{x-\xi}{\sigma}\right),\ k=0$	$\xi\leq x<\infty, k\leq 0$ $\xi\leq x<\xi+\frac{\sigma}{k}, k>0$	$k\ \epsilon R,\ \xi\ \epsilon R,\ \sigma>0$ k= shape parameter ξ =location parameter σ= scale parameter

5.1) Estimation of Parameter

- The parameters of the distribution have been estimated using the Maximum Likelihood Estimator (MLE) method
- Maximum likelihood is a popular technique for statistical inference because of its flexible and intuitive reasoning. The MLE is obtained as: -

$$L = f(x_1, x_2, \dots\dots, x_n|\theta) = f(x_1|\theta) * f(x_2|\theta) * \dots\dots.* f(x_n|\theta)$$

- The average log-likelihood function is easier to work with: -

$$\hat{l} = \frac{1}{n}\log L = \frac{1}{n}\sum_{i=1}^{n}\log f(x_i|\theta)$$

5.2) GOODNESS OF FIT TESTS

- Goodness of Fit Tests are statistical analyses used to determine how well a sample data set matches a distribution from a population.
- These tests assess if observed data deviate significantly from what was expected under a specified distribution.

Methods to Check Normality:

- Formal Tests: Statistical tests like Kolmogorov-Smirnov (K-S), Anderson-Darling (A-D), and Cramer-von-Mises.
- Graphical Methods: Visual tools like histograms, boxplots, and Quantile-Quantile (Q-Q) plots.

5.2) GOODNESS OF FIT TESTS

- Numerous methods are available to verify normality in statistical testing processes. Among them are tests known as Empirical Distribution Functions (EDF)
- EDF tests measure the difference between theoretical and actual distributions.
- Common EDF tests include Kolmogorov-Smirnov (K-S), Anderson-Darling (A-D), Cramer-von-Mises and Bayesian Information Criterion(BIC)

Kolmogorov–Smirnov (K-S) test:-

- Definition: The Kolmogorov-Smirnov (K-S) test is a non-parametric test used to compare a sample distribution with a reference probability distribution (one-sample K-S test) or to compare two sample distributions (two-sample K-S test).

5.2) GOODNESS OF FIT TESTS

Uses:

- Goodness of Fit: Assess if a sample comes from a specified distribution.
- Comparing Two Samples: Determine if two independent samples are drawn from the same distribution.
- Testing for Normality: Check if a sample follows a normal distribution or other theoretical distributions.
- Formula: The K-S statistic is calculated as:
  - $D_n = \max|F_x(x) - S_n(x)|$
- Where: Dn is the K-S statistic.
- Fx (x) is the empirical distribution function of the sample.
- Sn (x) is the cumulative distribution function of the reference distribution.
- A higher value of Dn indicates a larger discrepancy between the sample distribution and the reference distribution.

5.2) GOODNESS OF FIT TESTS

Anderson–Darling (A-D) test:-

- Definition: The Anderson-Darling (A-D) test is a statistical test used to determine if a sample comes from a specific distribution.

- It is known for its high efficacy compared to other Empirical Distribution Function (EDF) tests

- Uses:

Goodness of Fit: Evaluate if a sample fits a specified distribution, especially useful for normality tests.

Testing for Specific Distributions: More sensitive to deviations in the tails compared to the K-S test, making it suitable for detecting departures from normality or other specified distributions.

5.2) GOODNESS OF FIT TESTS

- **A-D Test Statistic:**

The quadratic class of EDF test statistics is expressed as A^2

- $A^2 = -\sum_{i=1}^{n}\{(2i-1)\ln F_X(x_i) + \ln[1 - F_X(x_{n+1-i})]\}\,/n\,] - n$

Where:

- A^2 is the Anderson-Darling test statistic.
- n is the sample size.
- x_i are the ordered data points.
- F_X is the cumulative distribution function of the specified distribution.
- ln is the natural logarithm
- A higher value of A^2 indicates a larger deviation from the specified distribution.

Cramer-von-Mises (CvM) test

- **Definition:** The Cramer-von-Mises is a criterion used for judging the goodness of fit of a cumulative distribution function F compared to a given empirical distribution function F*n, or for comparing two empirical distributions.
- It is also used as a part of other algorithms such as minimum distance estimation.

$$\omega^2 = \int_{-\infty}^{\infty} [F_n(x) - F^*(x)]^2 \, dF^*(x)$$

Formula:

- Let x_1,x_2,…..,x_n be the observed values, in increasing order. Then the test statistic is: -

$$T = \omega^2 = \frac{1}{12n} + \sum_{i=1}^{n} \left[\frac{(2i-1)}{2n} - F(x_i)\right]^2$$

5.2) GOODNESS OF FIT TESTS

Where:

ω^2 is the Cramer-von Mises test statistic.

n is the sample size.

x_i are the ordered data points.

F is the cumulative distribution function of the specified distribution.

A higher value of ω^2 indicates a larger deviation from the specified distribution.

5.2) GOODNESS OF FIT TESTS

Bayesian information criterion (BIC):-

Definition: The Bayesian Information Criterion (BIC) is a statistical measure used for model selection among a finite set of models. It helps to determine which model best fits the data by balancing model fit and complexity.

Uses:

- Model Selection: Identify the most appropriate model from a set of candidate models.
- Comparing Models: Compare models with different numbers of parameters and select the one that provides **the best trade-off between goodness of fit and model simplicity.**

- **Formula:**

$$BIC = kln\,(n) - 2\ln(\hat{L})$$

5.2) GOODNESS OF FIT TESTS

- Where

  $\hat{L}$ = the maximized value of the likelihood function of the model M, i.e. $\hat{L} = p(x \mid \hat{\theta}, M)$
- Where $\hat{\theta}$ are the parameter values that maximize the likelihood function and x is the observed data;
- n = the number of data points in x, the number of observations or equivalently, the sample size;
- k = the number of parameters estimated by the model

5.2) GOODNESS OF FIT TESTS

- **Quantile-Quantile(Q-Q) Plots:-**

Definition:

A Quantile-Quantile (Q-Q) plot is a graphical tool used to assess if a dataset follows a specified theoretical distribution by comparing their quantiles.

Construction:

- Step 1: Order the Data, Arrange the data points in ascending order to get the sample quantiles.
- Step 2: Determine Theoretical Quantiles ,Calculate the theoretical quantiles from the specified distribution.
- Step 3: Plot the Points

- Plot the sample quantiles against the theoretical quantiles on a scatter plot
- The observed data, X(i) and the estimated values, X(F) are displayed to create the Q-Q plot

Formula:

$$P_{i:n} = \frac{n-a}{N+1-2a}$$

- The majority of the plotting position formulas in use today use the following format: -
- Where,

n=1,2,3,…,N is the rank of the observed value X (with X(i) in ascending order).

N:Total number of observed values.

a: A constant used in calculating plotting positions.

- Plotting Positions: Blom's Plotting Position: Used for $N2$,$LN2$,$PIII3$,$LPIII3$and $PV3$ distributions.
- Constant: a=3/8
- Gringorten's Plotting Position: Used for GUM2,W2, and $GP3$distributions.
- Constant: a=0.44
- Cunnane's Plotting Position: Used for GEV distribution.
- Constant: a=0.4,a=0.4

Return Period:-

- The return period, also known as recurrence interval, is a statistical measure that estimates the average time interval between events of a certain intensity or size, such as floods, earthquakes, or storms.
- The determination of the recurrence interval, often known as the return period, is one of the main goals of frequency analysis.
- Uses:

Risk Assessment: Helps in assessing the likelihood and frequency of extreme events.

Infrastructure Design: Used in engineering to design structures like dams, bridges, and flood defenses to withstand events of a certain return period.

Environmental Planning: Aids in planning for natural disaster management and mitigation strategies.

Insurance: Assists in setting premiums and understanding risk exposure for natural disasters.

- Formula: If variable (x) is equal to or greater than an event size X_T and it happens once every T year, then P(X≥x) is the chance that the variable will occur in a given day.
- The return period T is calculated as :-

$$P(x \geq x_T) = \frac{1}{T}$$

$$T = \frac{1}{1 - P(x \leq x_T)}$$

- Interpretation:
- T = 100 years: Indicates that the event has a 1% chance (0.01 probability) of occurring in any given year.
- Higher Return Period: Indicates rarer events (e.g., 1000-year flood).
- Lower Return Period: Indicates more frequent events (e.g., 10-year storm).

6) RESULTS

1. The parameters of the distribution have been estimated using the Maximum Likelihood Estimator (MLE) in R software. Using our nine probability distributions that we have analyzed in the literature study, we calculated the values of the parameters for each of the 36 regions that were available to us, as indicated in the **Table 1**.
2. Our study's collected data revealed that each region's rainfall range differed significantly. The variations in rainfall can be attributed in part to the site's geographic position including its longitude latitude evaluation and surrounding environmental conditions.

- **Table 2** displays the Regions names, the findings of the statistical parameters mean, standard deviation, coefficient of skewness, minimum value, maximum value, medium and skewness for each of the regions that we've employed in our study. The absolute value of skewness reveals the relative magnitude of the deviations from the mean.

Descriptive Analysis

- The average for 14 regions lies within 0-300mm, for 16 regions it lies within 300mm-600mm, for 4 regions it lies within 600mm-900mm, for 2 regions lies it within 900mm-1200mm.

- The standard deviation for 33 regions falls between 0 - 200 mm, while for the other 3 regions it falls between 200mm-400mm. The amount of standard deviation with respect to its mean is shown by the coefficient of variation (CV), a relative measure of variability The coefficient to variation of all regions lies within 0-50mm.

- The coefficient of skewness of 4 stations are greater than 1, thus can be regarded as slightly skewed. 20 stations lie with the range of 0.5 to 1, thus highly-skewed. .

- While 12 stations are approximately symmetric, lying with the range of -0.5 to 0.5.

Best-Fit Results:

- The Regions name, best fit test statistic results of K-S, A-D, CVM and BIC are tabulated in the **Table 3**.
- Using several goodness of fit tests including K-S, AD, BIC and CVM tests we have computed a test statistic for each of the 36 regions.
- The best fit result is taken as the smallest goodness of fit results among all the probability distributions developed for each region.

- In order to determine which distribution is best suited for each region, a ranking system was created, with rank 1 being the best rank, rank 2 representing the second best and so on as shown in **Table 4**.

- **Table 4** displays regions names, distributions names , goodness of fit such as KS, AD, CVM AND BIC and their rankings and total rank is the sum of rank of each distribution. The distribution type with the lowest sum of the four rankings is selected as a best fit distribution for each of the stations.

- Seven stations had two distributions with the same value of best score (a tie between two types). They were LN2 and LPIII3 distributions in Gangetic West Bengal, LPIII3 and GUM2 distributions in Jammu and Kashmir and Ladakh, PIII3 and GEV3 distributions in Jharkhand, N2 and GEV3 distributions in Konkan and Goa, PIII3 and GEV3 distributions in Orissa, LN2 and GEV3 distributions in Rayalaseema, GEV3 and N2 distributions in Uttaranchal.

- In Assam and Meghalaya, the lowest rank is 6, shown by the GEV3 distribution, in Bihar the lowest rank is 4 shown by the GEV3 distribution, in Coastal Karnataka is 8 shown by GEV3 distribution. In Gujarat the lowest rank is 5 shown by GEV3 distribution.

- Table 4 show that just 2 regions i.e. 13 and 33 shower distinct distribution throughout all the 4 goodness of a test.

- Seven regions are best fitted by the same distribution over all four goodness affect tests.

- The regions names and distribution of lowest rank sums are displayed in Table 5, this table indicates the best fit distribution according to the lowest rank sum.

- From **table2** and **table 5** we can say that, among the lowest ranked distribution, all the regions with mean maximum monthly rainfall greater than 400 mm is best-fitted by either GEV3 or PIII3 distribution except region no. 22 which is best-fit by GUM2 distribution.

- All regions with skewness between 0 and 0.5 are symmetric and best fitted by PIII3 distribution except region number 14. GEV3 is the highest ranked-distribution in these regions. The regions with highly skewed distributions, best fitted by PIII3 and LPIII3 have skewness between 0.5-1.

- Among all the distributions it was found that in the best score result, 47% was best-fitted by GEV3 distribution, 20% was best fitted by PIII3 and LN2 distributions and 11% were best fitted by LPIII3 and GUM2 distributions, N2 distribution gave some best-fit results of 8%, W2 distribution gave only a few best-fit results of 3%.

- However, GP3 didn't gave any best-fit results. Based on Table 4 and 5, we can say that while PV3 is not among the lowest-ranking distributions in any region, it did provide some best fit results in areas such as Coastal Karnataka (shown by K-S and CVM test), Haryana, Chandigarh and Delhi (shown by K-S test) and West Rajasthan (shown by A-D test).

QQ-PLOT RESULTS

- By using a Q-Q plot, the level of fit on the extreme right tail can be studied. We have used the best-fit probability distributions for each region and using their parameters values, estimated using the Maximum Likelihood Estimator (MLE) fitted the Q-Q plot for each region.

- **The degree to which our selected model matches the gathered data is evident from the figure presented in the report.**

- Sub-Himalayan, Marathwada, Madhya Maharashtra, Lakshadweep, West Bengal and Sikkim, which yielded a GEV3 in all the four tests, is nearly a straight line indicating that the hypothesized distribution fits the data well.

- Based on the ranking score Orissa's two best suited models are GEV3 and PIII3. The two distributions offered an excellent match to the data.

- In certain cases, such as those in Nagaland, South Interior Karnataka, Manipur, Mizoram, and Tripura, Costal Andhra Pradesh, Kerala, South Interior Karnataka and Telangana, the fitted distributions frequently underestimated the extreme rainfall values. These cases included large deviations in the upper tails. There were other instances when severe rainfall levels were overstated, meaning that the real observed values fell below the predicted line. These examples include the under Andaman and Nicobar Island, Punjab, Himachal Pradesh, Jammu and Kashmir and Ladakh, Haryana, Chandigarh and Delhi. The bottom tail has either no deviations at all or very minor ones.

- There are other regions where the right tails appear to be fitted accurately by the presumed fitting distribution, including Chattisgarh, Lakshadweep, Sub-Himalayan West Bengal and Sikkim, Orissa and Uttaranchal. The data in regions like Chattisgarh and Sub-Himalayan West Bengal and Sikkim, Uttaranchal, Lakshadweep is best-fitted by the GEV3 distribution in the right tails.

Return Period Results

- According to the best fit probability model, we have then predicted return periods for 5 years, 10 years, 25years, 50 years ,100 years and 200 years. Table 6 shows the region names, their best fitted probability distribution along with the rainfall amounts for above specified return periods.

- The return levels for each region are shown here corresponding to maximum monthly rainfall in mm. For choosing the best fit probability distribution for return period corresponding to each division we evaluated, the best-fit results of K-S, A-D, CVM, BIC according to ranks, given in table 4, and chose the minimum ranked distribution for each of K-S, A-D, CVM, BIC results corresponding to each division

- We concluded that GEV3 fitted 32 regions, PIII3 fitted 16 regions, LPIII3 fitted 8 regions, GUM2 fitted 6 regions, N2 fitted 3 regions, PV3 fitted 2 distributions and W2 fitted only 1 region for return period estimation.

Return Period Results

- The information of return periods can help guide policy makers and planners. They can make their decisions for a given span of time using the expected maximum yearly rainfall as determined by the return period.

7) CONCLUSION:

- The rainfall patterns vary across all locations. The site's surroundings and geographic location have a big impact on how the rainfall pattern varies. We obtained the highest and lowest median, used standard deviation and skewness as a measure to divide our regions into categories.

- We have determined the best-fit probability distributions using lowest rank sum and using the four Goodness-of-fit tests such as K-S, AD, CVM and BIC to estimate the maximum annual rainfall for a given region. The maximum likelihood estimated was utilized to estimate the parameters for the distribution together with the graphical process called the Q-Q plot, the fits were determined using the K-S AD, CVM and BIC. Calculations were also made for the return period which is crucial in identifying the long-term concerns.

7) CONCLUSION:

- We **concluded that GEV3 fitted 32 regions, PIII3 fitted 16 regions, LPIII3 fitted 8 regions, GUM2 fitted 6 regions, N2 fitted 3 regions, PV3 fitted 2 distributions and W2 fitted only 1 region for return period estimation.**

- For most of the regions the testing statistics yielded different best-fit distribution types. Thus, a scoring system was used to determine the best bit distribution for each region. The best fit result of each region was taken as a distribution with the lowest sum of rank scores from each test statistics. GEV3, PIII3 and LPIII3 showed the larger number of best-fit results compared to other distributions. In the best score result 47% was best-fitted by GEV3 distribution, 20% was best fitted by each of the PIII3 and LN2 distributions and 11% were best fitted by each of LPIII3 and GUM2 distributions.

- The more practical result of this study was the return period calculation for all the regions and all distributions. The presentation of all distributions gave a clear inspection of the range of projected return period. Here 5-year, 10-year ,25-year ,50-year, 100-year and 200-year return periods of rainfall levels were illustrated.

- Better models of the danger and damage from floods and periods of heavy rainfall may be created using the findings of this study. These predictions can aid decision makers in developing programs that will ultimately save lives and property. Additionally, understanding the trend of really heavy rainfall would be helpful in many fields, including agriculture, construction, the environment and building architecture.

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